DO NOW — RECALL

A*: heuristic search I

Name: _____ Class: _____ Date: _____

1. Why are predecessors recorded when a distance improves?

2. Can Dijkstra assume negative edge weights are safe?

3. What is 2+3+2?

4. Starting S=0, edge S–A=4, then A–G=2. What is the S→A→G path cost?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

They allow reconstruction of the actual shortest route.

2. Model answer

No. Its shortest-path guarantee requires non-negative weights.

3. Model answer

7.

4. Model answer

6.

Vocabulary — know these terms

g(n) — Known best cost from the start to vertex n.

h(n) — Estimated cost from vertex n to the goal.

f(n) — Combined score g(n)+h(n).

frontier / open set — Discovered vertices awaiting expansion.

admissible heuristic — An estimate that never exceeds the true cheapest remaining cost.

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Update the table

From N, O improves to g4,h1,f5.

From O, G improves to g6,h0,f6.

Select G and reconstruct M→N→O→G, total cost 6.

IMPROVE: Explain one correction to your own reasoning.

NEW LEARNING

Read, annotate and explain

g, h and f have different meanings

g is distance already travelled; h estimates the remaining distance; f=g+h is the estimated total.

Start with a frontier

From S, A gets g2,h3,f5, while B gets g6,h1,f7. Choose A, not simply the smallest h.

Improve a discovered route

From A, B improves to g4,h1,f5. From B, G improves to g6,h0,f6.

Stop on the selected goal

When G is selected from the frontier, reconstruct S→A→B→G with total cost 6.

Keep the heuristic safe

The estimates h(S)=5,h(A)=3,h(B)=1,h(G)=0 never exceed the actual cheapest remaining cost.

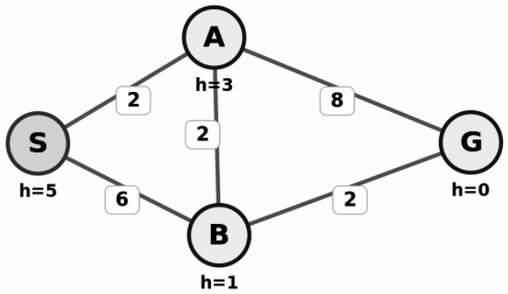
Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning

START AT S



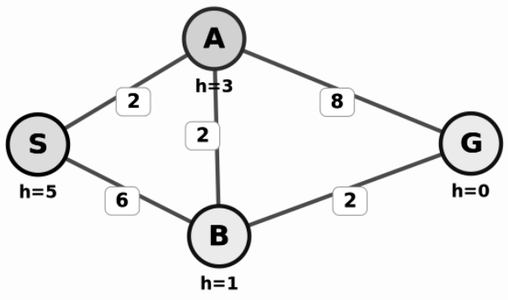
Node	g	h	f
A	∞	3	∞
B	∞	1	∞
G	∞	0	∞
S	0	5	5

S: g=0, h=5, f=5

START AT S A*: S: g=0, h=5, f=5

New Learning

EXPAND S



Node	g	h	f
A	2	3	5
B	6	1	7
G	∞	0	∞
S	0	5	5

A: f=2+3=5; B: f=6+1=7

EXPAND S A*: A: f=2+3=5; B: f=6+1=7

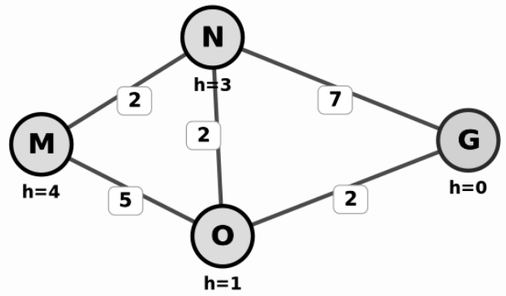
We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace A* on M, N, O, G using the supplied estimates.

From M, choose N because f(N)=2+3=5 < f(O)=5+1=6.

REACH G



Node	g	h	f
G	6	0	6
M	0	4	4
N	2	3	5
O	4	1	5

M→N→O→G = 6

REACH G A*: M→N→O→G = 6

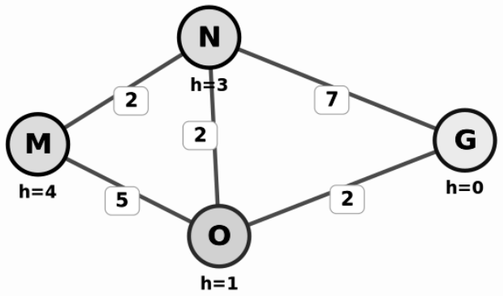
We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace A* on M, N, O, G using the supplied estimates.

From M, choose N because f(N)=2+3=5 < f(O)=5+1=6.

EXPAND N



Node	g	h	f
G	9	0	9
M	0	4	4
N	2	3	5
O	4	1	5

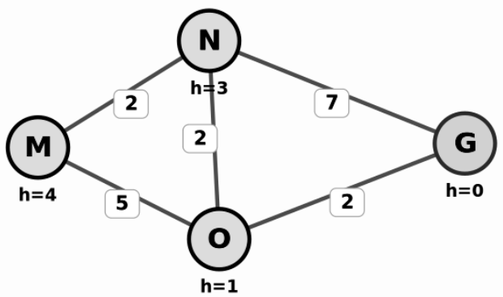
O improves: g=4,f=5

EXPAND N A*: O improves: g=4,f=5

Trace A* on M, N, O, G using the supplied estimates.

From M, choose N because f(N)=2+3=5 < f(O)=5+1=6.

EXPAND O



Node	g	h	f
G	6	0	6
M	0	4	4
N	2	3	5
O	4	1	5

G improves: g=6,f=6

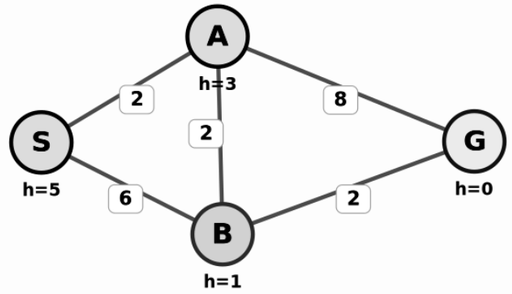
EXPAND O A*: G improves: g=6,f=6

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning

EXPAND A



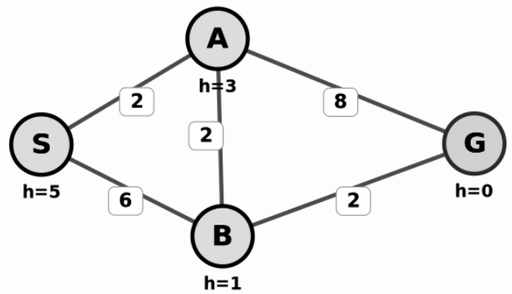
Node	g	h	f
A	2	3	5
B	4	1	5
G	10	0	10
S	0	5	5

B improved: g=4, f=5

EXPAND A A*: B improved: g=4, f=5

New Learning

EXPAND B



Node	g	h	f
A	2	3	5
B	4	1	5
G	6	0	6
S	0	5	5

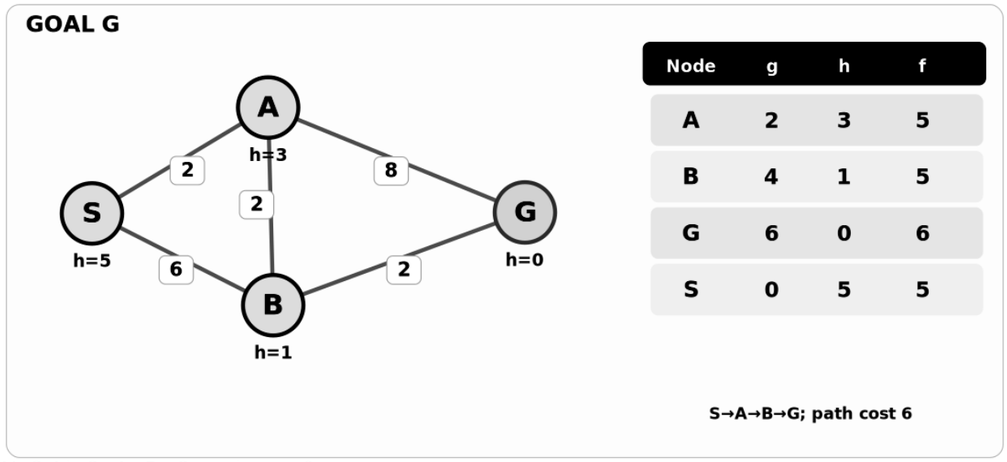
G improved: g=6, f=6

EXPAND B A*: G improved: g=6, f=6

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



GOAL G A*: S→A→B→G; path cost 6

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A* chooses the next vertex solely by its smallest heuristic h(n).

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Explain why admissibility matters. For the graph S–A=2, A–G=4 and S–G=10, what is the greatest admissible h(A), and why?

I DO — WORKED EXAMPLE

Follow the worked method

Trace A* on P, Q, R, G using the supplied admissible estimates.

Predict the next state

From P, Q has g3+h3=f6, R has g6+h2=f8. Choose Q.

Update the table

Through Q, R improves to g4,h2,f6; G gets g11.

Update the table

From R, G improves to g6,h0,f6.

Update the table

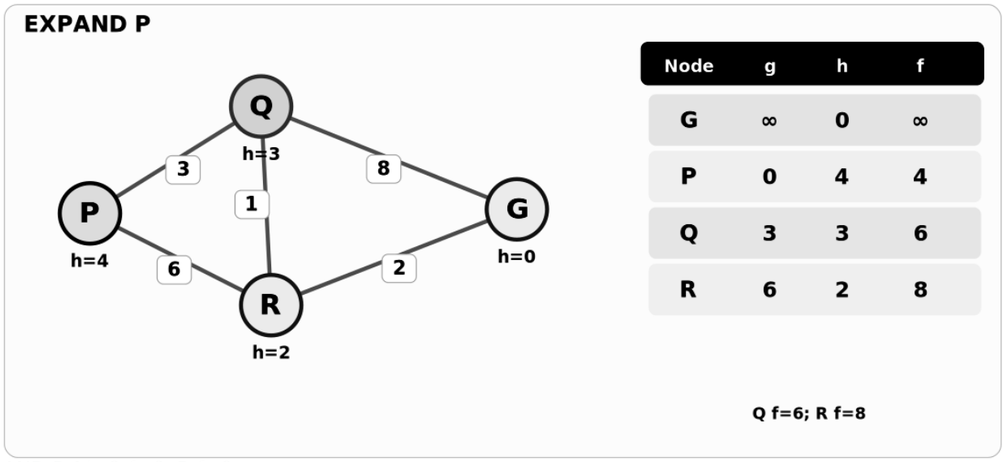
Select G: predecessor chain yields P→Q→R→G, cost 6.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace A* on P, Q, R, G using the supplied admissible estimates.

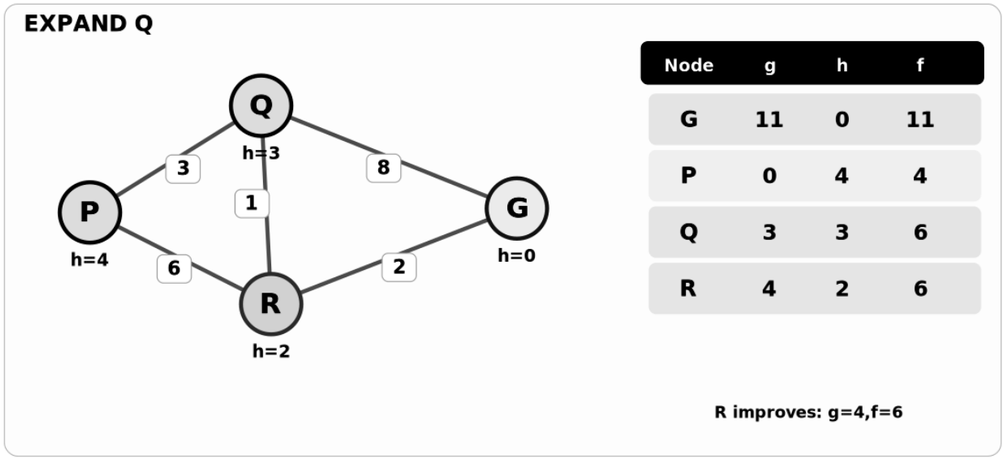
From P, Q has g3+h3=f6, R has g6+h2=f8. Choose Q.



EXPAND P A*: Q f=6; R f=8

Trace A* on P, Q, R, G using the supplied admissible estimates.

From P, Q has g3+h3=f6, R has g6+h2=f8. Choose Q.



EXPAND Q A*: R improves: g=4,f=6

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

A node has g=7 and h=3. Give f and define each value.

Question 2

Graph edges S–A=2, S–B=5, A–B=1 and B–G=2. Heuristics: h(S)=4,h(A)=2,h(B)=1,h(G)=0. Calculate f for A and B when first found and state the optimal S→G route.

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict the next state From M, choose N because f(N)=2+3=5 < f(O)=5+1=6.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

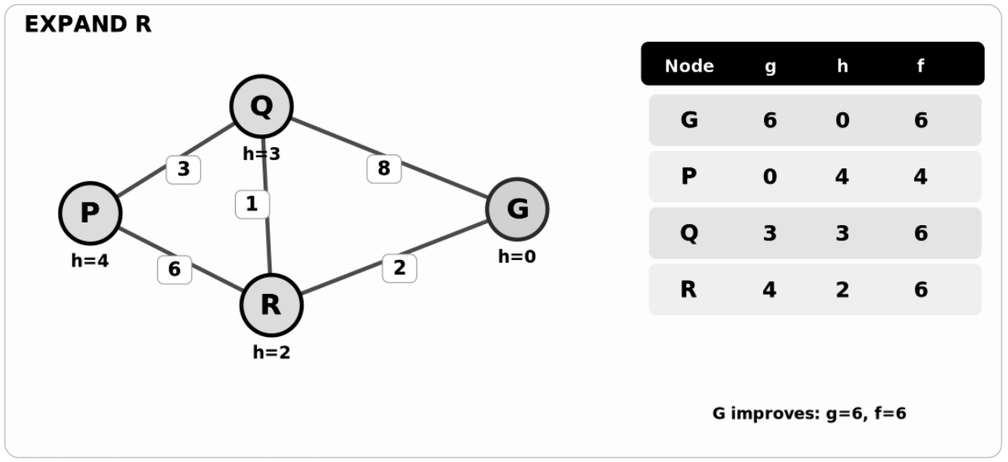
IMPROVE — Write one precise correction or stronger explanation in your own answer.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace A* on P, Q, R, G using the supplied admissible estimates.

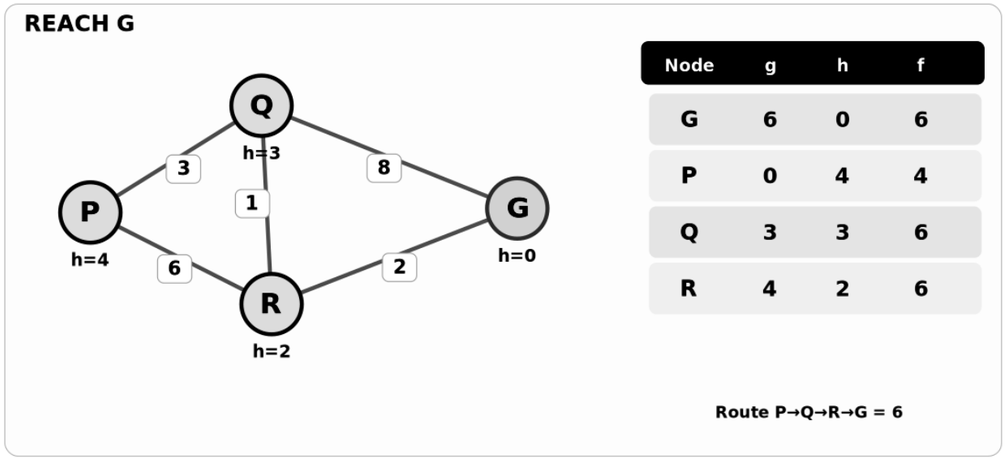
From P, Q has g3+h3=f6, R has g6+h2=f8. Choose Q.



EXPAND R A*: G improves: g=6, f=6

Trace A* on P, Q, R, G using the supplied admissible estimates.

From P, Q has g3+h3=f6, R has g6+h2=f8. Choose Q.



REACH G A*: Route P→Q→R→G = 6

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

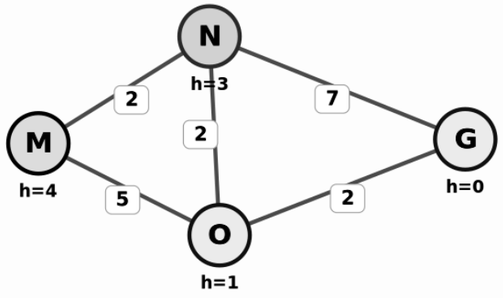
We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace A* on M, N, O, G using the supplied estimates.

From M, choose N because f(N)=2+3=5 < f(O)=5+1=6.

EXPAND M



Node	g	h	f
G	∞	0	∞
M	0	4	4
N	2	3	5
O	5	1	6

N f=5; O f=6

EXPAND M A*: N f=5; O f=6